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AIM & OBJECTIVE

AIM: To design and implement a smart EV charging station system using ESP32 that detects battery connection, monitors charging progress, displays battery percentage and charging time, and calculates charging cost automatically.

- OBJECTIVE:**
- To detect EV battery presence using a voltage sensor.
 - To automatically switch load between shop load and EV charging using a relay.
 - To monitor battery percentage during charging.
 - To calculate charging cost based on time.



INTRODUCTION

- The rapid growth of electric vehicles (EVs) has increased the demand for efficient, economical, and sustainable charging solutions.
- Solar-powered EV charging systems are emerging as an effective approach to reduce dependence on conventional electricity and minimize operating costs.
- Integrating solar energy with smart monitoring and load management enhances system efficiency and ensures reliable charging
- This project proposes the renewable solar energy to charge electric vehicles while intelligently managing available power.



LITERATURE SURVEY

S. No.	Author Name	Title of the Paper/ Publication Details	Inference
1	M. Singh, B. Kumar et al., IEEE Transactions on Power Electronics	“Design of Solar Powered EV Charging Station with Battery Storage,” IEEE, 2019.	Supports the proposed system’s use of a secondary battery for stable power supply and efficient energy management.
2	P. Verma, R. Gupta et al., International Journal of Renewable Energy Research.	“Smart Energy Management for Standalone Solar EV Charging System,” IJ RER, 2020.	Validates the relay-based automatic switching between shop load and battery charging used in the proposed model.
3	K. Lee, J. Park et al., IEEE Access	“IoT-Enabled Monitoring System for Electric Vehicle Charging Stations,” IEEE Access, 2021.	Supports the LCD display of timer, battery percentage, and cost calculation in the proposed system.



EXISTING SYSTEM

- EV charging is mostly grid-dependent.
- No solar integration in small shops.
- High electricity cost during peak hours.
- No energy storage for backup.
- No prepaid billing or energy-based charging.
- Shop owners cannot use EV charging energy for shop loads.

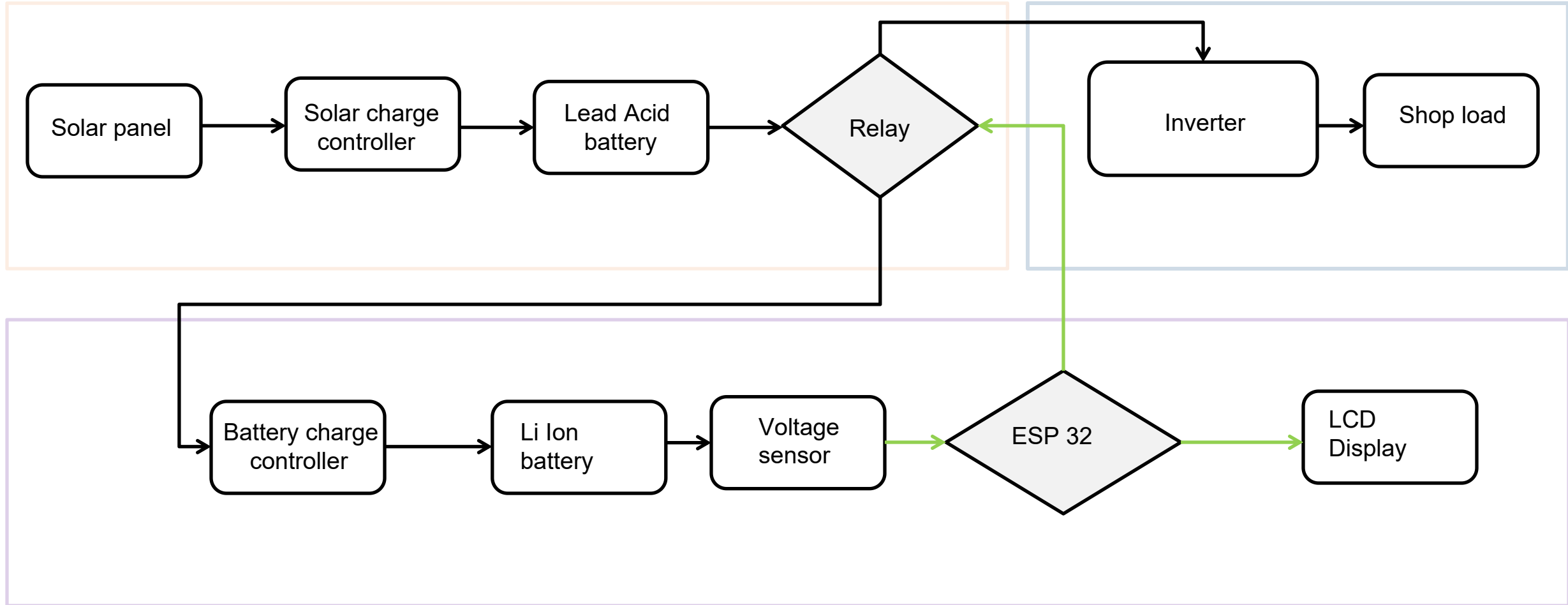


PROPOSED SYSTEM(contd.,)

- Solar rooftop-based EV charging system for shop parking.
- Low-cost secondary battery storage.
- Bidirectional power flow.
- Prepaid EV charging using energy-based billing.
- ESP32 for monitoring, control, and automation.
- Voltage & current sensing with display.
- Automatic relay control for charging and load sharing.
- Low-cost prototype, scalable for real-world use.



PROPOSED BLOCK DIAGRAM





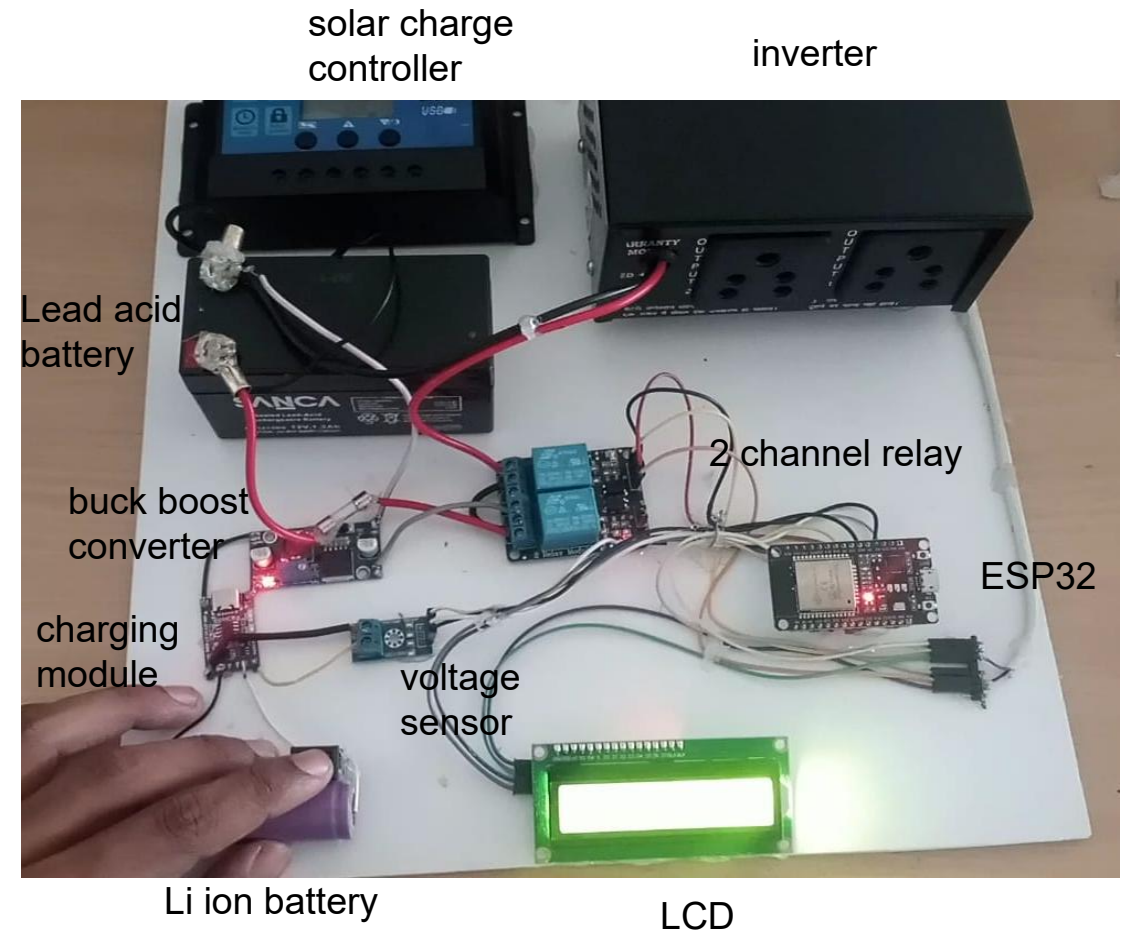
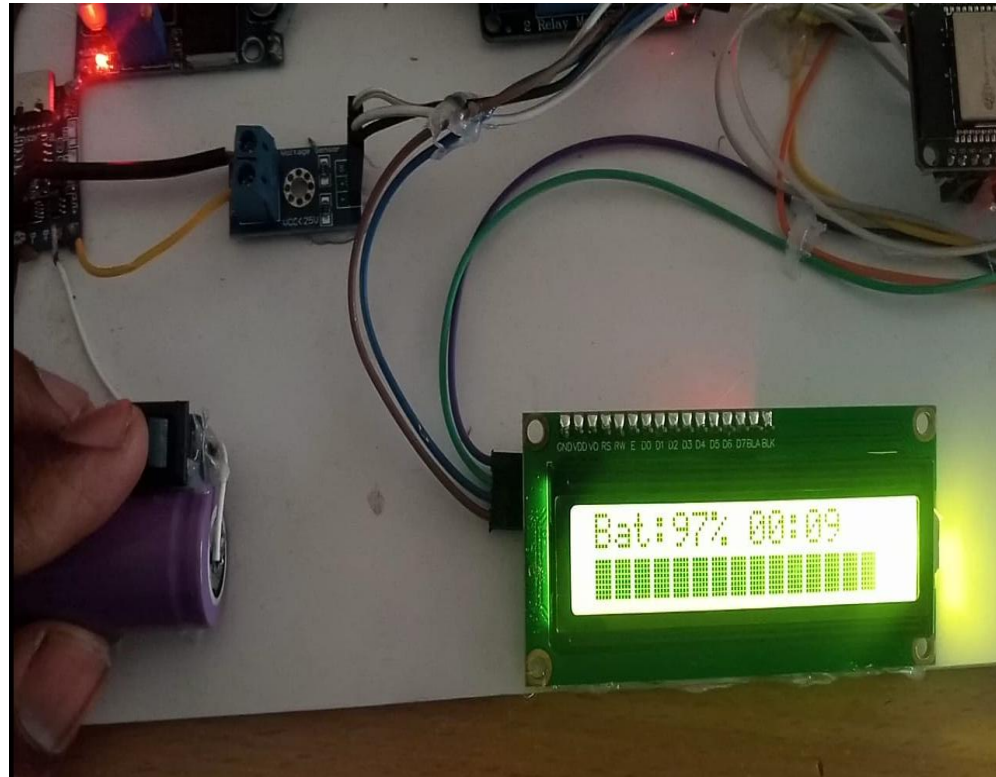
DESIGN CONSIDERATION

The following factors were considered:

- Voltage compatibility: ESP32 ADC limit is 3.3 V
- Common grounding: Required for stable readings
- Relay isolation: To safely switch loads
- Noise filtering: Needed for stable battery readings
- Power efficiency: TP4056 used for Li-ion safety
- User interface clarity: 16×2 LCD chosen for simplicity
- Cost effectiveness: Low-cost components used
- Scalability: System can be extended to IoT



HARDWARE RESULTS





CONCLUSION & FUTURE SCOPE

- The Solar EV Charging System using ESP32 was successfully developed to detect battery connection, monitor charging, calculate cost, and switch loads automatically. The system provides a reliable and cost-effective solution for small-scale EV charging.
- IoT cloud monitoring
- Mobile app integration
- RFID payment system
- Fast charging support
- Multiple vehicle ports



ADVANTAGES & APPLICATIONS

- Uses renewable solar energy
- Reduces electricity cost
- Automatic load switching
- Real-time monitoring & cost calculation
- Battery backup support
- Low maintenance and scalable
- Homes and apartments
- Small shops and businesses
- Parking areas
- Colleges and campuses
- EV charging stations
- Office buildings



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Thank you